

Units of volume and capacity, and converting between them

Cambridge insert: the cube of the length factor, and the litre that sits between the two systems.

LESSON 30

1 × 40 MIN

GEOMETRY 11, §14 (EXTENSION)

IGCSE E13.1–13.4

LESSON CLOCK

10'

12'

12'

8'

3'

Cubing the factor	10 min
Litres	12 min
Choosing a unit	12 min
Density	8 min
Homework	3 min

LEARNING OBJECTIVES

- Convert between mm^3 , cm^3 and m^3 .
- Convert between millilitres, litres and cubic units.
- Choose an appropriate unit for a given object.
- Use density to link volume and mass.

TEXTBOOK

National	pp. 153–156
Cambridge	Core & Extended, pp. 40–44
Workbook	IGCSE Exercise 13.1

01

Explanation

Cube the length factor

THE ONE RULE

If lengths convert by k , volumes convert by k^3 . $1\text{ m} = 100\text{ cm}$, so $1\text{ m}^3 = 100^3 = 1\,000\,000\text{ cm}^3$.

CONVERSION	LENGTH FACTOR	VOLUME FACTOR
cm → mm	10	1000
m → cm	100	1 000 000
km → m	1000	10 ⁹
m → mm	1000	10 ⁹

THE COMMONEST CONVERSION ERROR IN THE SYLLABUS

Writing $1\text{ m}^3 = 100\text{ cm}^3$. Areas need k^2 and volumes k^3 ; only lengths use k itself.

Litres

$$1\text{ ml} = 1\text{ cm}^3 \quad 1\text{ litre} = 1000\text{ cm}^3 \quad 1\text{ m}^3 = 1000\text{ litres}$$

The litre is defined as a cubic decimetre — a cube of side 10 cm. That single fact generates every line above, and it is why a 1 m³ tank holds exactly a tonne of water.

OBJECT	TYPICAL CAPACITY
a teaspoon	5 ml
a can of drink	330 ml
a bucket	10 litres
a bath	150 litres
a small water tank	1 m³ = 1000 litres

Choosing a unit, and density

An answer of 0.0000034 m³ or 34 000 000 mm³ is correct and useless. Choose the unit that makes the number sit between about 0.1 and 1000.

$$\text{density } \rho = \frac{m}{V} \text{ so } m = \rho V$$

Water has $\rho = 1\text{ g/cm}^3$, which is why 1 litre of water weighs 1 kg and 1 m³ weighs 1 tonne. Steel is about 7.8 g/cm³, pine about 0.5.

Example. A steel bar is a cylinder $r = 2\text{ cm}$, $h = 50\text{ cm}$. Its volume is $200\pi \approx 628\text{ cm}^3$, so its mass is about $628 \times 7.8 \approx 4900\text{ g} \approx 4.9\text{ kg}$.

02 Worked examples**EXAMPLE 1** Convert 2.5 m^3 to cm^3 and to litres.

① $1 \text{ m}^3 = 1\,000\,000 \text{ cm}^3$.

② $2\,500\,000 \text{ cm}^3$.

③ $\div 1000$ gives 2500 litres.

ANSWER

$$2\,500\,000 \text{ cm}^3 = 2500 \text{ litres}$$

EXAMPLE 2 A tank is $1.2 \times 0.8 \times 0.5 \text{ m}$. How many litres does it hold?

① $V = 0.48 \text{ m}^3$.

② $\times 1000$

③ $= 480 \text{ litres}$.

ANSWER

$$480 \text{ litres}$$

EXAMPLE 3 A steel cylinder has $r = 2$ cm and $h = 50$ cm. Find its mass at 7.8 g/cm³.

① $V = \pi \times 4 \times 50 = 200\pi \approx 628.3$

② $m = 628.3 \times 7.8$

③ ≈ 4901 g.

ANSWER

≈ 4.9 kg

03 Interactive model

Show a litre bottle beside a 10 cm cube — they hold the same.

Converting volume

1 m³ in cm³:

100

10 000

1 000 000

1000

1 cm³ in mm³:

10

100

1000

10 000

1 litre in cm³:

100

1000

10 000

1

1 m³ in litres:

100

1000

10 000

1 000 000

0.5 m³ in litres:

50

500

5000

5

1 litre of water weighs:

100 g

1 kg

10 kg

1 g

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Cubic centimetre	Kub santimetr	Кубический сантиметр
Cubic metre	Kub metr	Кубический метр
Millilitre	Millilitr	Миллилитр
Litre	Litr	Литр
Capacity	Sig'im	Вместимость
Conversion factor	O'tkazish koeffitsienti	Коэффициент перевода
Density	Zichlik	Плотность
Mass	Massa	Масса
Appropriate unit	Mos birlik	Подходящая единица

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Volumes convert by:

k

k^2

k^3

$3k$

1 ml equals:

1 mm³

1 cm³

10 cm³

1 litre

3 m³ in litres:

300

3000

30 000

3

Density is:

$$mV$$

$$\frac{m}{V}$$

$$\frac{V}{m}$$

$$m + V$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. 1 m^3 in cm^3

ANSWER $1\,000\,000$

2. 1 cm^3 in mm^3

ANSWER 1000

3. 2 litres in cm^3

ANSWER 2000

4. 5000 cm^3 in litres

ANSWER 5

5. 0.25 m^3 in litres

ANSWER 250

6. 750 ml in cm^3

ANSWER 750

7. 1 litre of water in kg

ANSWER 1

Medium · the standard question

7 PROBLEMS

1. 2.5 m^3 in cm^3 and litres

ANSWER $2\,500\,000 \text{ cm}^3$, 2500 L

2. Tank $1.2 \times 0.8 \times 0.5 \text{ m}$ in litres

ANSWER 480

3. A cylinder $r = 5$, $h = 20 \text{ cm}$ in litres

ANSWER ≈ 1.57

4. A steel cylinder $r = 2$, $h = 50 \text{ cm}$ at 7.8 g/cm^3 : mass

ANSWER $\approx 4.9 \text{ kg}$

5. $3\,500\,000 \text{ mm}^3$ in litres

ANSWER 3.5

6. A bath of 150 litres in m^3

ANSWER 0.15

7. A block of pine $20 \times 10 \times 5 \text{ cm}$ at 0.5 g/cm^3 : mass

ANSWER 500 g

Hard · stretch and reason

7 PROBLEMS

1. A pool $25 \times 10 \times 1.8$ m: capacity in litres

ANSWER 450 000

2. A pipe of length 5 m, $R = 6$, $r = 5$ cm: metal mass at 7.8 g/cm^3

ANSWER $\approx 135 \text{ kg}$

3. A raindrop of radius 2 mm: volume in ml

ANSWER ≈ 0.0335

4. A tonne of water occupies how many m^3 ?

ANSWER 1

5. A gold bar $10 \times 5 \times 3$ cm at 19.3 g/cm^3 : mass

ANSWER $\approx 2.9 \text{ kg}$

6. Explain why 1 m^3 is not 100 cm^3

ANSWER Volumes need the cube of the length factor

7. A tank holds 2400 litres and is 2×1.5 m in base. Find its depth

ANSWER 0.8 m

07 Homework

Homework — 4 tasks

Choose the unit that keeps the number readable, and say why.

- Convert 4.2 m^3 to cm^3 and to litres, and 6500 mm^3 to cm^3 .
- A tank measures $2 \times 1.5 \times 0.9$ m. Find its capacity in litres.
- A steel pipe 4 m long has outer radius 7 cm and inner radius 6 cm. Find the mass of metal at 7.8 g/cm^3 .

4. Explain in three sentences why volumes convert with the cube of the length factor, with one worked example.